

Nanoarchitectonics of High-Performance and Flexible n-Type Organic–Inorganic Composite Thermoelectric Fibers for Wearable Electronics

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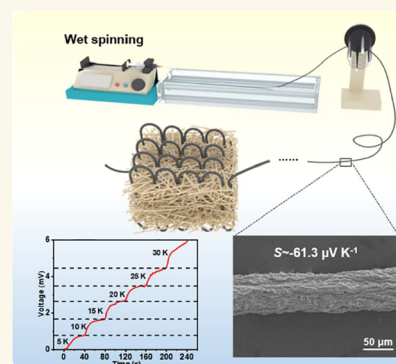
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ABSTRACT: Since most conductive polymers are *p*-type, developing high-performance *n*-type organic–inorganic composite thermoelectric (TE) fibers is a great challenge. Herein, poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS)-coated Ag_2Te nanowires (PC- Ag_2Te NWs) were prepared by a liquid-phase reaction using PEDOT:PSS-coated Te nanowires (PC-Te NWs) as templates, and the PEDOT:PSS/PC- Ag_2Te NWs composite fibers were then prepared by wet spinning. As the content of PC- Ag_2Te NWs increases, the composite fiber changes from *p*-type to *n*-type. The PEDOT:PSS coating greatly improves the dispersibility of Ag_2Te NWs in the PEDOT:PSS matrix, resulting in an ultrahigh content of 87.5 wt % of PC- Ag_2Te NWs in the composite fibers, which exhibited a Seebeck coefficient of $-61.3 \mu\text{V K}^{-1}$ and a power factor of $65.3 \mu\text{W m}^{-1} \text{K}^{-2}$. The power factor value is higher than those of previously reported *n*-type composite TE fibers. Contrary to the estimated thermal conductivity in other reports, in this work, the thermal conductivity of the composite fibers was measured via a transient photoelectrothermal (TPET) technique. In addition, the composite fiber has good tensile properties and mechanical strength, elongating at a break of 47.37% and a tensile stress of 6.59 MPa. For an application demonstration, a self-powered temperature sensor was assembled, which can utilize the vertical temperature difference between the human body and the environment and respond quickly to a small temperature difference.

KEYWORDS: PEDOT:PSS, Ag_2Te nanowire, thermoelectric fiber, flexible, wet spinning



INTRODUCTION

As thermoelectric (TE) materials can realize solid-state energy conversion between heat and electricity, which could play a role in solving the current problems of energy shortage and environmental pollution, TE materials have received significant attention in the last decades.^{1–4} Flexible TE materials can convert the temperature difference (ΔT) between the human body and the environment into electricity based on the Seebeck effect, which shows a broad application prospect in the field of smart wearables.⁵ To evaluate the energy conversion efficiency of TE materials, a dimensionless figure of merit (ZT) is introduced, which is usually quantified by $ZT = S^2\sigma T/\kappa$, where S , σ , κ , and T denote the Seebeck coefficient, electrical conductivity, thermal conductivity, and absolute temperature, respectively.⁶ $S^2\sigma$ is also known as the power factor (PF), which is commonly used to estimate the properties of TE films or fibers.⁷

Compared to bulk materials or films based on elastomeric substrates, TE fibers have high structural designability and are

ideally suited for integration into garments or fabrics.^{8–11}

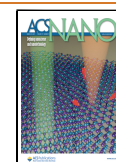
Organic TE fibers are mainly conductive polymers, which usually have the advantages of low toxicity, lightweight, and good flexibility.¹² For example, the continuous wet-spun *p*-type poly(3,4-ethylenedioxythiophene): poly(styrenesulfonate) (PEDOT:PSS) fibers post-treated with sulfuric acid (H_2SO_4) had a PF of $147.8 \mu\text{W m}^{-1} \text{K}^{-2}$ (with $\sigma = 4029.5 \text{ S cm}^{-1}$ and $S = 19.2 \mu\text{V K}^{-1}$) at room temperature (RT).¹³ TE fibers based on a conducting polymer have higher flexibility than inorganic TE fibers, but their TE performance is much lower than that of the latter. Inorganic TE nanomaterials can be added to the conducting polymer matrix to improve the TE properties while

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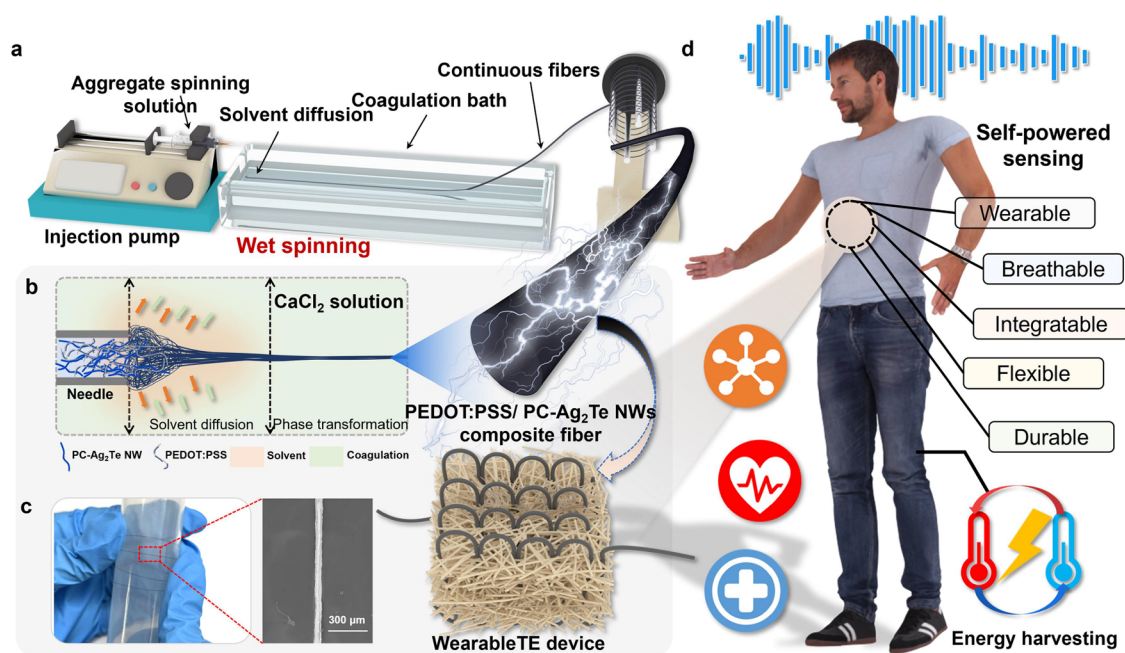


Figure 1. (a) Schematic plot demonstrating the preparation process of the PEDOT:PSS/PC-Ag₂Te NWs composite fibers and wearable TE device. (b) Schematic diagram of wet spinning mechanism. (c) Digital photograph and SEM image of the composite fiber. (d) Schematic of the application of wearable TE devices based on the composite fibers.

maintaining the flexibility of the conducting polymer.^{14,15} For example, Xu et al.¹⁶ mixed high *S* Te nanowires (NWs) with PEDOT:PSS and prepared PEDOT:PSS/Te composite fibers with an *S* value of 32 $\mu\text{V K}^{-1}$ by wet spinning. However, the content of Te NWs in the composite fibers can only be maintained at a low level of ~ 30 wt % due to the significant difference in chemical properties between the organic and inorganic phases. Recently, our group developed an in situ synthesis method to improve the compatibility between organic and inorganic phases. For example, when synthesizing inorganic NWs, a several-nanometer-thick organic (PEDOT:PSS,^{17,18} polypyrrole (PPy),¹⁹ or polyvinylpyrrolidone (PVP)²⁰) layer is coated in situ. This type of organic-coated inorganic NWs can be compatible with the organic matrix, greatly increasing its content in the organic matrix. Most recently, we in situ synthesized PEDOT:PSS coated Te NWs (PC-Te NWs) and wet-spun PEDOT:PSS/PC-Te NWs composite fibers. The PC-Te NWs can be oriented in the composite fibers with a content of 65 wt %.²¹ The composite fiber after appropriate post-treatment showed an ultrahigh *S* of 121.2 $\mu\text{V K}^{-1}$ and a PF of 385.4 $\mu\text{W m}^{-1} \text{K}^{-2}$ at RT.

High-conversion efficiency TE devices usually consist of *p*-type and *n*-type TE materials with matching properties.²² In order to prepare flexible TE fibers, conductive polymers are usually used. Because conductive polymers are mainly *p*-type, most reported flexible TE fibers are *p*-type. Until now, only a few *n*-type TE composite fibers have been reported, and most of them have poor stability and low TE properties.^{23,24} For example, a stretchable fiber constructed by wet spinning based on the composite of 2D Ti₃C₂T_x MXene nanosheets and polyurethane (PU) has a *S* of only $-8.3 \mu\text{V K}^{-1}$ and a PF of 0.086 $\mu\text{W m}^{-1} \text{K}^{-2}$; highly conductive *n*-type polymer fibers obtained from *n*-doped poly(3,7-dihydrobenzo[1,2-b:4,5-b']-difuran-2,6-dione) (*n*-PBDF) by wet-spinning have a *S* of only $-19 \mu\text{V K}^{-1}$; polyethylenimine (PEI)-doped single-walled carbon nanotubes (SWCNTs) fiber possesses a *S* of -24.96

$\mu\text{V K}^{-1}$ and a PF of 38.78 $\mu\text{W m}^{-1} \text{K}^{-2}$; the *ISI* of which decreased by $\sim 15\%$ after 14 days under ambient condition. To obtain *n*-type composite fibers, one approach is to use *n*-type conductive polymers as the matrix, while another requires the *n*-type component (such as *n*-type inorganic TE nanostructures) to be present in sufficiently high concentrations to counteract the influence of the *p*-type conductive polymer. However, the compatibility between the inorganic and organic phases is usually poor, making the fabrication of stable and high-performance *n*-type composite TE fibers a significant challenge.

Ag₂Te is an *n*-type TE material with the characteristics of “phonon-liquid electron-crystal” and superelectronic conductivity, which has potential applications in waste heat power generation.^{28–30} The Ag₂Te bulk was reported to have an *S* value of $-129 \mu\text{V K}^{-1}$ and a σ value of about 600 S cm^{-1} .³¹ PEDOT:PSS-coated Ag₂Te (PC-Ag₂Te) nanorods were prepared by a wet-chemical method and flexible PC-Ag₂Te nanocomposite films were prepared on poly(ether sulfone) (PES) substrates through vacuum filtration and mechanical pressing processes with an *S* value of $-62.3 \mu\text{V K}^{-1}$.³² Therefore, it is believed that Ag₂Te has the potential to be employed as an inorganic TE filler for the preparation of *n*-type composite fibers in combination with PEDOT:PSS.

In this work, PC-Te NWs were synthesized in situ via a hydrothermal reaction, then PC-Ag₂Te NWs were prepared using PC-Te NWs as templates by a liquid-phase reaction, and finally, a series of PEDOT:PSS/PC-Ag₂Te NWs composite fibers with different contents of the PC-Ag₂Te NWs were wet spun. The PEDOT:PSS layer improves the dispersion of Ag₂Te NWs in the PEDOT:PSS matrix, allowing the content of PC-Ag₂Te NWs to reach 87.5 wt %. Thus, *n*-type composite fibers with high *S* ($-61.3 \mu\text{V K}^{-1}$) and high PF (65.3 $\mu\text{W m}^{-1} \text{K}^{-2}$) were obtained. The κ of the composite fiber was 1.05 $\text{W m}^{-1} \text{K}^{-1}$ measured by using a transient photon-electrothermal (TPET) technique, and the ZT value was calculated to be

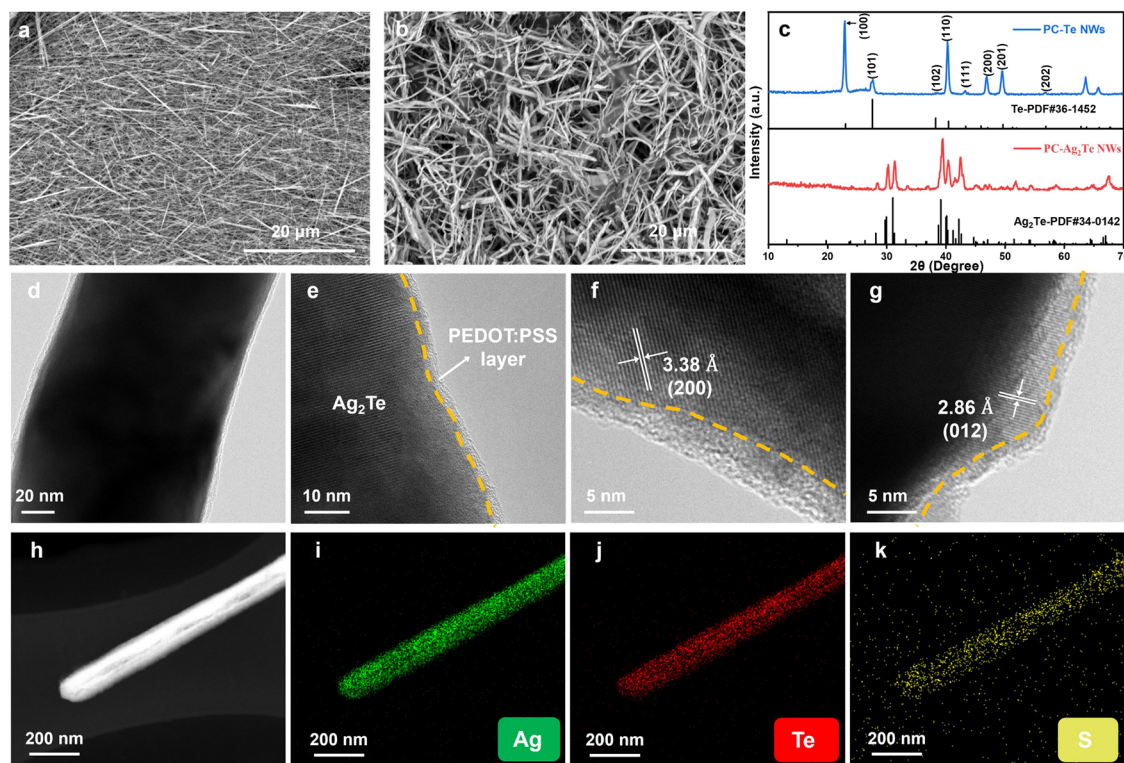


Figure 2. SEM images of the PC-Te NWs (a) and PC-Ag₂Te NWs (b); (c) X-ray diffraction patterns of the PC-Te NWs and PC-Ag₂Te NWs. (d–g) TEM and HRTEM images of the PC-Ag₂Te NWs; HAADF (h) and EDS-STEM elemental mapping of Ag (i), Te (j), and S (k) of the PC-Ag₂Te NW.

~0.02. The interfacial effect of PEDOT:PSS/Ag₂Te and the carrier transport characteristics are also discussed. In addition, a self-powered flexible temperature sensor was assembled with the composite fibers, and its sensitive and stable sensing capability demonstrated the potential for application in wearable electronics.

RESULTS AND DISCUSSION

Design of the PEDOT:PSS/PC-Ag₂Te NWs Composite Fibers. The schematic of the preparation of high-performance and flexible PEDOT:PSS/PC-Ag₂Te NWs composite fibers and the application of a wearable TE device assembled is shown in Figure 1. Ag₂Te with a negative *S* value was chosen as a filler to prepare the *n*-type composite TE fibers. To improve the filler's dispersion, content, and stability in PEDOT:PSS aqueous solution, PC-Ag₂Te NWs were first synthesized using PC-Te NWs as templates. The PC-Te NWs were prepared by in situ coating a layer of PEDOT:PSS on the surface of Te NWs during the hydrothermal process. The length of the PC-Te NWs is several tens of micrometers, as can be seen from the SEM image (Figure 2a), and their morphology is similar to that of the Te NWs shown in Figure S2. The high aspect ratio can be attributed to the hexagonal crystal structure of Te and the structure-directing role of PVP. PC-Te NWs have a 2–3 nm thick amorphous PEDOT:PSS nanolayer on the surface of the Te NWs. The XRD pattern of the PC-Te NWs is shown in Figure 2c, and all XRD peaks can be well correlated with the standardized data of hexagonal Te (JCPDS card No. 36-1452). Notably, abnormal strong (100) and weak (101) peaks are observed in the XRD pattern of PC-Te NWs, suggesting that PC-Te NWs have a preferential growth direction along [001], which is the same as reported in ref 33.

PC-Ag₂Te NWs were obtained from the reaction between PC-Te NWs and AgNO₃ in EG. XRD analysis of the PC-Ag₂Te NWs reveals that all peaks can be indexed to the monoclinic Ag₂Te phase (JCPDS Card No. 34-0142). Due to the crystal structural difference between the hexagonal Te and the monoclinic Ag₂Te, the resulting Ag₂Te NWs, shown in Figure S3, exhibit significantly reduced lengths. In contrast, the PC-Ag₂Te NWs derived from PC-Te NWs maintain relatively high aspect ratios, although they become curved (Figure 2b). This should be due to the protective effect of the PEDOT:PSS layer during the reaction. Ascorbic acid is commonly employed as a reducing agent to convert Te to Ag₂Te. The significant difference in crystal structure between Te and Ag₂Te leads to the formation of very short Ag₂Te NWs under vigorous reaction conditions, as shown in Figure S4. These shortened NWs are unsuitable for fabricating composite fibers. In the XRD patterns of both the PC-Te NWs and PC-Ag₂Te NWs, the diffraction peak corresponding to the interchain planar ring stacking distance *d* (010) of PEDOT ($2\theta = 25.5^\circ$) is not obvious.³³ It was speculated to be the partial detachment and/or thinning of the PEDOT:PSS layer due to constant stirring during the prolonged liquid-phase reaction, coupled with its amorphous nature. The interface between Ag₂Te and the PEDOT:PSS layer can be observed. The morphology of the PC-Ag₂Te NWs is shown in the TEM and HRTEM images (Figure 2d–g), and the PEDOT:PSS layer is about 2–5 nm thick. Moreover, lattice spacings of 0.338 and 0.286 nm can be observed in the HRTEM images of Figure 2f,g, corresponding to the (200) and (012) crystal planes of Ag₂Te, respectively, indicating good crystallinity. Figure 2h–k show that in addition to Ag and Te elements, the S element, which should be from PEDOT:PSS, is also uniformly distributed in the PC-

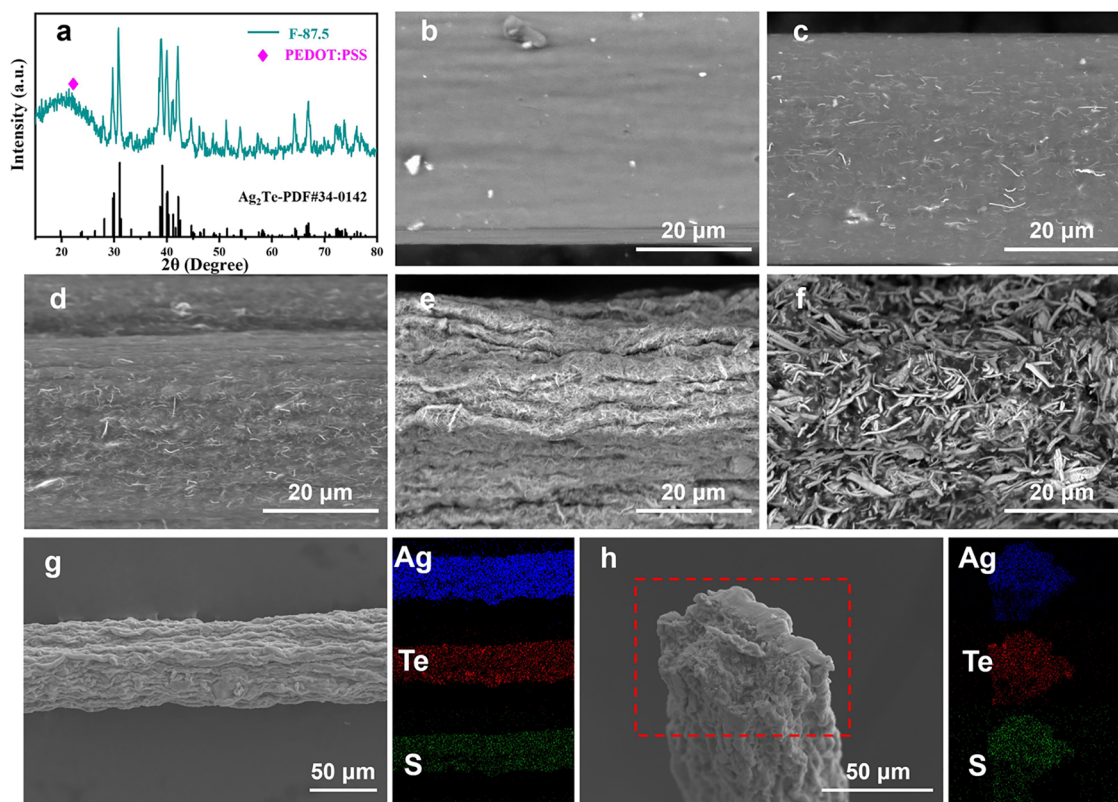


Figure 3. (a) X-ray diffraction pattern of the F-87.5. SEM images of the F-0 (b), F-50 (c), F-85 (d), F-87.5 (e), and F-90 (f), respectively. HAADF and EDS-STEM elemental mapping of Ag, Te, and S of the surface (g) and cross-section (h) of the F-87.5.

Ag₂Te NW, indicating good encapsulation of the NW by PEDOT:PSS, and this favors the compatibility between PC-Ag₂Te NWs and PEDOT:PSS matrix.

Although the *S* value of a single PC-Ag₂Te NW is difficult to measure, the *S* of a PC-Ag₂Te NWs film prepared by ultrasonically dispersed in ethanol and deposited on a porous nylon membrane through vacuum-assisted filtration is measured to be $-91.39 \mu\text{V K}^{-1}$. PEDOT:PSS is a *p*-type organic material with a *S* of $19 \mu\text{V K}^{-1}$.³⁴ Therefore, the *S* value of the composite fiber is determined by the *S* and content of these two phases. In a simple binary system, if the constituent interfaces are ignored, then series or parallel models are usually used to estimate the TE performance parameters of the composite. For the composite fiber, the conventional series and parallel model between the PC-Ag₂Te NWs and PEDOT:PSS can be written as³⁵:

$$S_{c,s} = \frac{V_1 \kappa_2 S_1 + V_2 \kappa_1 S_2}{x_1 \kappa_2 + x_2 \kappa_1} \quad (1)$$

$$S_{c,p} = \frac{V_1 \sigma_1 S_1 + V_2 \sigma_2 S_2}{x_1 \sigma_1 + x_2 \sigma_2} \quad (2)$$

where $S_{c,s}$, $S_{c,p}$, S_1 , and S_2 are the *S* of series-connected composite, parallel-connected composite, PC-Ag₂Te NWs, and PEDOT:PSS matrix, respectively. Additionally, V_1 and V_2 denote the volume fractions of PC-Ag₂Te NWs and the PEDOT:PSS matrix, respectively. As shown in Figure S5, only when the volume fraction of the filler is greater than 10 vol %, the composite fibers show *n*-type with appreciable negative *S*. It was converted to mass fractions as listed in Table S1, thus four groups of composite fibers with PC-Ag₂Te NWs mass

fractions ≥ 50 wt % were prepared by wet spinning. For comparison, pure PEDOT:PSS fibers were also prepared.

In the XRD pattern of F-87.5, in addition to the diffraction peaks of Ag₂Te, the diffraction peaks of the PEDOT:PSS matrix are also present (Figure 3a). To further investigate the structure, Fourier transform infrared (FT-IR) spectroscopy was conducted on PEDOT:PSS, PC-Ag₂Te NWs, and the composite fibers (F-87.5), and the results are shown as Figure S6, where the peaks at ~ 1900 to $\sim 2300 \text{ cm}^{-1}$ correspond to the testing substrate. For PEDOT:PSS, the quinoid structure and aromatic C=C stretching vibrations are found at $1584\text{--}1514 \text{ cm}^{-1}$, the C=C bonds at 1352 cm^{-1} , and the C-S bond vibrations of the thiophene ring at $822\text{--}670 \text{ cm}^{-1}$. The peak at 1046 cm^{-1} corresponds to the S-C phenyl bonds in sulfonic acid. Ag₂Te is an inorganic compound, and its infrared vibration absorption is much weaker compared with PEDOT:PSS; hence, its FTIR signals are not observed. Therefore, the FTIR spectrum of F-87.5 is essentially consistent with that of PEDOT:PSS. In the FTIR spectrum of PC-Ag₂Te NWs, the characteristic peaks of the PEDOT:PSS layer can be observed; however, the peaks are slightly weaker due to the low content of PEDOT:PSS. SEM images of the five-type fibers at different magnifications are shown in Figures 3b–f and S7. From the low magnification images, it can be seen that F-0, F-50, F-85, and F-87.5 are relatively homogeneous with diameters of 53.5, 41.9, 45.7, and 52.2 μm , respectively, whereas F-90 is uneven in diameter. As seen from the high-magnification images, the surface of F-0 is very smooth. The addition of PC-Ag₂Te NWs changes the rheological properties of the spinning solution, and the surface of the composite fibers remains smooth when the content of PC-Ag₂Te NWs is low, which is attributed to the adhesive

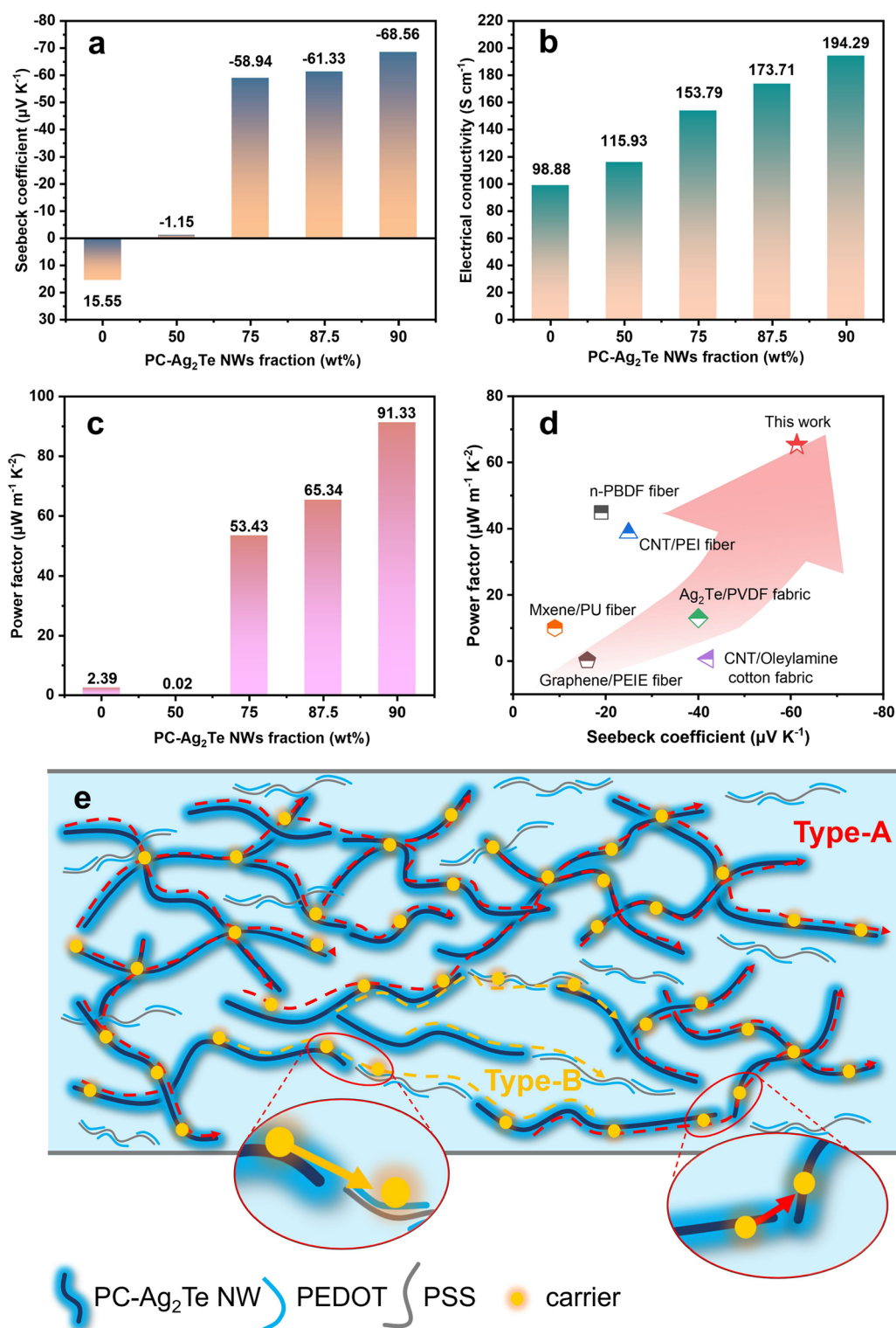


Figure 4. TE performance of the composite fibers with different PC-Ag₂Te NWs contents at RT. (a) Seebeck coefficient, (b) electrical conductivity, and (c) power factor. (d) Comparison of power factor of the F-87.5 and other reported PEDOT:PSS-based composite fibers and fabrics (PVDF refers to polyvinylidene fluoride; PEIE refers to polyethylenimine ethoxylated). (e) Schematic illustration of carrier transport mechanisms in the composite fiber.

effect of PEDOT:PSS that confines the PC-Ag₂Te NWs in the fibers.³⁶ However, when the content of the PC-Ag₂Te NWs reaches 87.5 wt %, the surface of the composite fiber becomes rough and many wrinkles appear. Nevertheless, F-87.5 can still be continuously spun, as shown in Figure S8a. However, when

the content of the PC-Ag₂Te NWs reaches 90 wt %, only intermittent short fibers can be spun (Figure S8b). The mass fraction of inorganic filler in this work reaches 87.5 wt %, which is significantly higher than that of other reported PEDOT:PSS-based composite fibers. On the one hand, it is

due to the high density of Ag₂Te itself, and on the other hand, it is because of the PEDOT:PSS layer, resulting in the Ag₂Te NW's better dispersibility and stability in the PEDOT:PSS matrix. This in situ coating method can be extended to other polymer-based composite fibers, offering a promising strategy for improving the performance and stability of various composite materials by enhancing the compatibility and dispersion of inorganic fillers in organic matrices. When the PC-Ag₂Te NW content is 90 wt %, it exceeds the PEDOT:PSS matrix's accommodation limit, making it difficult to obtain continuous and long fibers. The surface morphology of the wet spun PEDOT:PSS/Ag₂Te NWs composite fiber (with <50 wt % Ag₂Te NWs) from the mixed solution of PEDOT:PSS and Ag₂Te NWs is displayed in Figure S9, in which the distribution of Ag₂Te NWs is inhomogeneous. Moreover, the composite fibers fabricated directly using Ag₂Te NWs are brittle when the Ag₂Te NW content reaches 50 wt %, whereas the surface morphology of the PEDOT:PSS/PC-Ag₂Te NWs composite fibers is much more homogeneous, indicating that our novel experimental design is very successful. Figure 3g–h show the Ag, Te, and S elemental mapping images at the surface and cross-section of the F-87.5. It can be found that Ag, Te, and S elements are uniformly and densely distributed in the composite fiber. In F-0, the S element is uniformly distributed, and Ag and Te elements are absent (Figure S10).

TE Properties of PEDOT:PSS/PC-Ag₂Te NW Composite Fibers. Since the diameter of a single fiber is too small to establish a ΔT value between the two ends, 10 fibers were aggregated into a bundle and its S and σ at RT were measured with Cryoall CTA according to ref 37 and the result is shown in Figure 4a,b. The cross-sectional SEM image of the F-87.5 fiber bundle is shown in Figure S11, and the cross-sectional areas of all fiber bundles are listed in Table S2. 50 wt % of PC-Ag₂Te NWs is required to counteract the positive S of PEDOT:PSS, resulting in an S value of $-1.2 \mu\text{V K}^{-1}$ for F-50. If an n -type matrix, such as n-PBDF, is selected, then a higher S can be obtained with less PC-Ag₂Te NWs content. When the contents of PC-Ag₂Te NWs are 85, 87.5, and 90 wt %, the S of the composite fibers are -58.9 , -61.3 , and $-68.6 \mu\text{V K}^{-1}$, respectively. The PEDOT:PSS fibers, i.e., F-0, exhibit an σ of only 98.9 S cm^{-1} . The σ value of the composite fibers gradually increases with the addition of the PC-Ag₂Te NWs, and the σ of F-50, F-85, F-87.5, and F-90 are 115.9, 153.8, 173.7, and 194.3 S cm^{-1} , respectively. To explain the σ characteristics of the PEDOT:PSS/PC-Ag₂Te NWs composite fibers, series, and parallel models are also applied:

$$\sigma_{c,s}^{-1} = V_1\sigma_1^{-1} + V_2\sigma_2^{-1} \quad (3)$$

$$\sigma_{c,p} = V_1\sigma_1 + V_2\sigma_2 \quad (4)$$

where $\sigma_{c,s}$, $\sigma_{c,p}$, σ_1 , and σ_2 are the σ of series-connected composites, parallel-connected composites, the PC-Ag₂Te NWs, and PEDOT:PSS matrix, respectively. As shown in Figure S12, the values of experimentally measured σ lie between those predicted by the series and parallel models, and as the content of the PC-Ag₂Te NWs increases, the σ of the composite fibers gets closer to the result of the parallel model. It is hypothesized that this is due to carriers being transported mainly through the more conductive PC-Ag₂Te NWs when the content of PC-Ag₂Te NWs is close to the limit and the PEDOT:PSS matrix primarily acts as a bonding agent. The PF of F-90 was calculated to be the largest (Figure 4c), but it could not be continuously spun. Hence, F-87.5 with a PF of

$65.3 \mu\text{W m}^{-1} \text{ K}^{-2}$ was selected for subsequent studies. Owing to the tunability of the σ of PEDOT:PSS, the post-treatment of F-87.5 with H₂SO₄ or EG, respectively, was also explored, resulting in a substantial increase in the σ , yet the $|S|$ was reduced to $\sim 10 \mu\text{V K}^{-1}$. That is because when the σ of the PEDOT:PSS matrix is increased dramatically after the post-treatment, the carrier transport pathway changes from the PC-Ag₂Te NWs to the PEDOT:PSS matrix, and thus, the p -type conduct shows an overwhelming advantage. Therefore, such post-treatment is not good for this composite fiber. Figure 4d shows that F-87.5 has the highest $|S|$ and the highest PF compared with most reported n -type composite TE fibers or fabrics.^{25–27,38–40} The microstructure and electron transport schematic of the composite fibers are shown in Figures 4e and S13. The PEDOT:PSS nanolayer results in a uniform distribution of PC-Ag₂Te NWs, forming a three-dimensional PC-Ag₂Te NW electrical network. According to the electrical transport properties (Figure 4a), it is deduced that when the content of PC-Ag₂Te NWs is low in composite fiber, like in F-50, the weight of carrier transport through the PEDOT:PSS matrix and the PC-Ag₂Te NWs network is almost equal, leading to the S being close to zero. When the content of PC-Ag₂Te NWs is high, like in F-87.5, most of the NWs are interconnected, and carriers mainly transport in the NWs network through numerous Ag₂Te/PEDOT:PSS heterojunctions. In this case, the TE properties of the composite fiber are mainly determined by the PC-Ag₂Te NWs network. This transport mechanism is termed Type A. Of course, carrier transport may be also achieved through the bridging of NWs with the PEDOT:PSS molecule chains from the matrix, such as Type B shown in Figure 4e. Namely, in the composite fiber, carrier transport involves numerous Ag₂Te/PEDOT:PSS heterojunctions.

As measuring the Hall effect on fiber-like materials is a great challenge, we could not provide experimental proof of the energy filtering effect. However, by comparing our results with previous studies, we found that the trend of the S of the composite fibers (Figure S14) agrees with that observed in PEDOT:PSS/Te NWs composite fibers. In ref 34, the authors highlighted the role of the energy filtering effect. Additionally, Hall measurements performed by our group on PEDOT:PSS/Ag₂Se/CuAgSe (PC-CuAg₂Se₂) and Ag₂Se/CuAgSe (CuAg₂Se₂) composite films demonstrated that the carrier concentration of the PC-CuAg₂Se₂ is lower than that of the CuAg₂Se₂ film, while the carrier mobility of the PC-CuAg₂Se₂ is higher, suggesting there exists energy-filtering effect.¹⁸ Besides, Choi et al.⁴¹ also considered that there are energy filtering effects at the reduced graphene oxide (rGO)/PEDOT:PSS and PEDOT:PSS/Te NW heterojunctions in rGO/PEDOT:PSS/Te NWs composite films. We used Series and parallel models, which describe two extreme cases that do not account for interfaces, to estimate the Seebeck coefficient of the composite fibers and compared the results with experimental values (see Figure S15). When the content of PC-Ag₂Te NWs is high, the experimental $|S|$ surpasses the calculated values, suggesting the contribution of the energy filtering effect. According to previously reported works on composite films^{18,41} and composite fiber,³⁴ and combining our experimental data, we deduce that in the composite fibers, there is an energy filtering effect at the PEDOT:PSS/Ag₂Te heterojunction: low-energy carriers are blocked while high-energy carriers pass the junction, enhancing carrier mobility, the S , and overall TE performance.

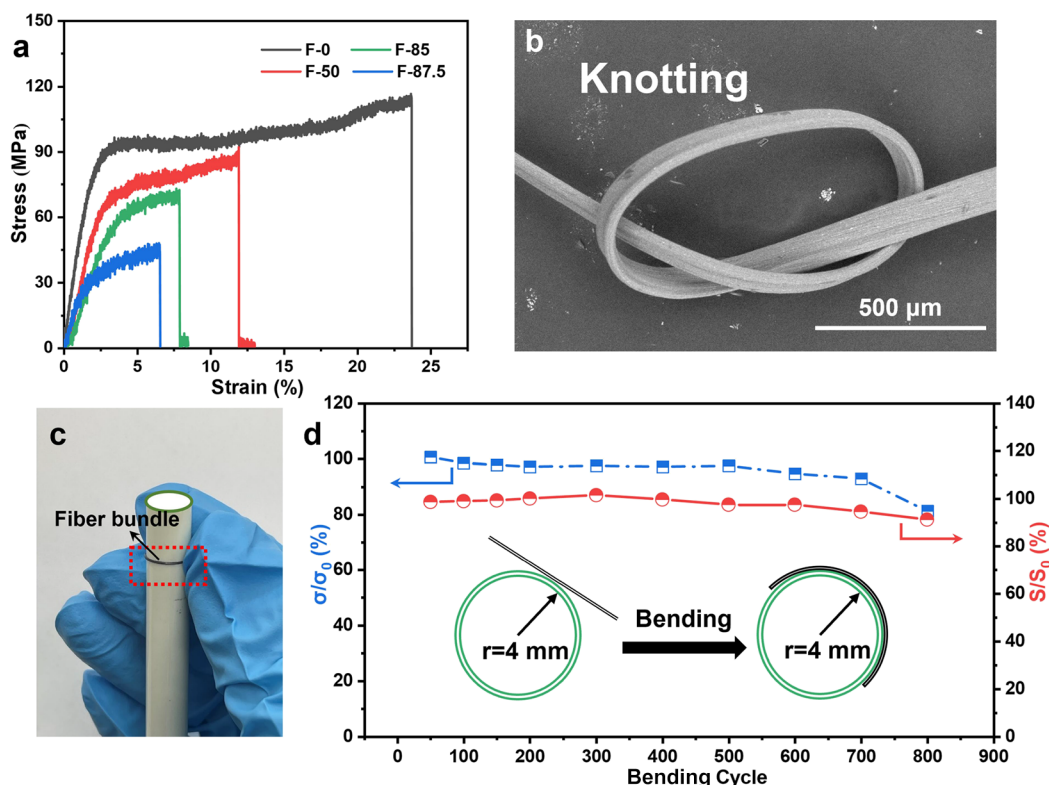


Figure 5. (a) Stress–strain curves of the composite fibers with different content of PC-Ag₂Te NWs. (b) SEM image of a knotted composite fiber. (c) Digital photo showing a fiber bundle bent around a rod with a radius of 4 mm. (d) Flexibility test result of the fiber bundle.

Furthermore, based on the above analysis, the composite fibers are believed to have low κ for the following reasons: (1) a large number of interfaces can effectively scatter phonons and (2) PEDOT:PSS has intrinsically low κ .³⁴ To verify this speculation, the TPET technique⁴² was used to measure the κ of F-0, F-50, and F-87.5, as listed in Table S3. Among them, F-0 has the lowest κ of 0.21 W m^{−1} K^{−1}. The κ of the composite fibers increases with the increase of the PC-Ag₂Te NWs content, and F-87.5 has a κ value of 1.05 W m^{−1} K^{−1} with a calculated ZT value of 0.02. In previous reports, the κ and ZT values of the composite fibers are generally estimated. This is the first time that the κ of *n*-type PEDOT:PSS/inorganic composite fibers and the corresponding ZT value are reported.

Mechanical Properties and Air Stability of the PEDOT:PSS/PC-Ag₂Te NW Composite Fibers. For the application of wearable electronics, the PEDOT:PSS/PC-Ag₂Te NWs composite fibers are also expected to have good mechanical strength and flexibility. Figure 5a shows the stress–strain curves of the composite fibers with the PC-Ag₂Te NWs ranging from 0 to 87.5 wt %. The tensile strength of F-0 is 116.30 MPa, and the elongation at a break is 23.74%. The tensile strength and elongation at the break of the composite fiber decrease with the increase of the PC-Ag₂Te NW content, which is ascribed to two reasons: (1) Ag₂Te NWs are intrinsically brittle, and (2) the PC-Ag₂Te NWs affect the continuity of the PEDOT:PSS matrix when the PC-Ag₂Te NW content is high. The tensile strength of F-87.5 is 47.37 MPa, and the elongation at a break is 6.59%. The Supporting Video shows that F-87.5 can easily withstand a weight of 10 g, which proves that it can meet some mild deformation and stretching requirements. Figure 5b shows that F-87.5 can be easily tied into a knot with a diameter of ~1 mm. Figure 5c,d illustrate that the σ and S decrease by 18.73 and 8.75%,

respectively, after a bundle of F-87.5 bent 800 times around a circular rod with a 4 mm radius. Figure S16 demonstrates the folding ability of the bundle. The good flexibility of the composite fibers is demonstrated, which is derived from (1) the intrinsic high flexibility of PEDOT:PSS, (2) the thickening effect of PEG, and (3) the strong bonding between the PEDOT:PSS and PC-Ag₂Te NWs.

One of the common problems of *n*-type organic–inorganic composite TE fibers is their environmental stability, so we performed XPS analysis on the Ag₂Te NWs, PC-Ag₂Te NWs, and F-87.5 as shown in Figure S17. The peaks of Ag 3d_{5/2}, Ag 3d_{3/2}, and Te 3d_{5/2}, Te 3d_{3/2} at binding energies of 368.3, 374.3, 572.5, and 582.9 eV were detected in all the samples, which indicates that the valence states of Ag and Te are (+1) and (−2), respectively. It is noteworthy that the binding energies at 586.6 and 576.1 eV in the high-resolution XPS spectra of Te 3d in Ag₂Te NWs correspond to the Te(IV)3d state, which is absent in these of the PC-Ag₂Te NWs, and F-87.5. This suggests that the Te atoms near the surface of Ag₂Te NWs have a low coordination number and thus are easily oxidized in air, while the PEDOT:PSS nanolayer can protect the Ag₂Te NWs from oxidation. In addition, we observed that S remained at 91.65% after F-87.5 was exposed to air for 20 days (Figure S18), indicating good stability.

Self-Powered Temperature Sensors. The composite fiber has the advantages of high TE performance, good flexibility, and high seamability and can be easily integrated into some conventional fabrics. As shown in Figure 6a, we have designed a TE device consisting of a long bundle of composite fibers sewn into a nonwoven fabric, and the bundle is intermittently coated with the silver paste. As shown in Figure S19, the pattern of the device is variable by a rational structural design. Two Peltier components are used to establish and

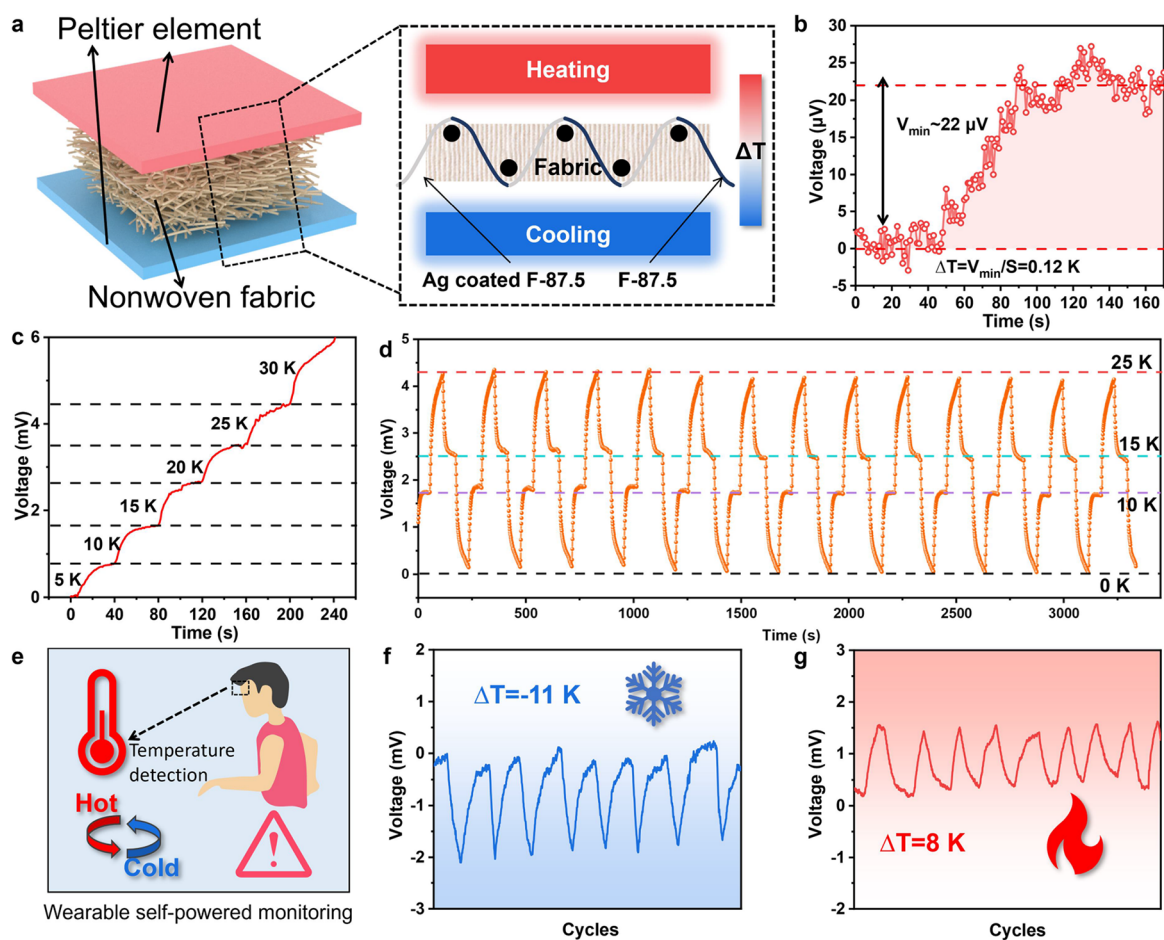


Figure 6. (a) Schematic diagram of the operating principle of the TE device. (b) Minimum detection temperature of the TE device. (c) Voltage response of the TE device to different ΔT . (d) Voltage response of the TE device to multiple cycles at ΔT . (e) Schematic of the TE device being used as a wearable self-powered temperature sensor. Voltage response of the TE device in periodic vicinity to (f) a cold object and (g) a hot object.

regulate the ΔT between the two ends of the device, which can operate as a self-powered temperature sensor. The sensor detected the lowest ΔT signal of 0.12 K (Figure 6b), demonstrating the high-temperature resolution accuracy. The sensor's recognition of different temperature signals was also characterized. When applying the variable ΔT at the two ends of the device, the output thermal voltage showed a regular increase. (Figures 6c and S20). To demonstrate its durability and reliability, a ΔT program "0–10–25–15 K" was designed using the Peltier components as shown in Figure 6d, and the temperature sensor operated stably over multiple cycles. In addition, the sensor has good wearability and can be integrated into fabrics for everyday wear for self-powered sensing (Figure 6e). The sensor can accurately differentiate and warn of (Figure 6f) cold and (Figure 6g) hot objects. Therefore, the composite fiber developed in this work has a very promising application in self-powered wearable smart devices.

CONCLUSIONS

In conclusion, high-performance *n*-type PEDOT:PSS/PC-Ag₂Te NWs composite fibers were prepared by wet spinning for the first time. The TE parameters of the composite fibers were theoretically analyzed using the series/parallel model and practically tested, and the microstructure and carrier transport characteristics were also investigated. The composite fiber

containing 87.5 wt % PC-Ag₂Te NWs exhibited an *S* value of $-61.3 \mu\text{V K}^{-1}$, a PF value of $65.3 \mu\text{W m}^{-1} \text{K}^{-2}$, and good flexibility and tensile properties, which can be integrated into fabrics for practical applications. The self-powered temperature sensor assembled from F-87.5 and nonwoven fabrics can effectively utilize the vertical ΔT between the environment and the heat source, demonstrating ultrahigh sensitivity (capable of detecting a minimum ΔT of 0.12 K). Furthermore, the exceptional TE performance of the composite fibers highlights their potential for fabricating highly flexible, lightweight, and efficient TE generators. This high-performance *n*-type composite fiber combines flexibility, breathability, and power generation capabilities, making it a highly promising candidate for wearable electronic devices.

EXPERIMENTAL SECTION

The raw materials and experimental details are given in Note S1. As shown in Figure S1, PC-Te NWs were synthesized according to the reported hydrothermal method,²¹ based on which PC-Ag₂Te NWs were obtained by a liquid phase reaction. Finally, a series of PEDOT:PSS/PC-Ag₂Te NWs composite fibers were prepared by continuous wet spinning. To investigate the effect of the content of PC-Ag₂Te NWs on the TE properties, the samples with PC-Ag₂Te NWs content of 0, 50, 85, 87.5, and 90 wt % were named F-0, F-50, F-85, F-87.5, and F-90, respectively.

The assembly process of the self-powered temperature sensor is described in Note S2, and the various characterization studies and

measurements of the PC-Ag₂Te NWs, PEDOT:PSS/PC-Ag₂Te NWs composite fibers, and self-powered temperature sensor are given in Note S3.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsnano.5c01168>.

Detailed experimental information and other characterization results (PDF)

F-87.5 can easily withstand a weight of 10 g (MP4)

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Author Contributions

J.L.: writing – original draft, methodology, investigation, conceptualization. X.H.: imaging processing, design and testing of the device. J.W., S.Z., M.Z., C.W., and G. D.: investigation and measurement. R.L.: resource. L.W.: resource and methodology. L.C.: review and editing. K.C.: writing – review and editing, resources, methodology, conceptualization. J.L. and X.H. are contributed equally to this work.

Notes

The authors declare no competing financial interest.

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